Ammar Hafez

SOFTWARE DEVELOPMENT | C++ PROGRAMMING | TECHNICAL PROJECTS

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PROFILE

Computer Game Development student with practical programming experience through university coursework and independent projects. Work spans C++ development, algorithms, computer systems, and interactive applications. Interested in software engineering, tools programming, and game development opportunities.

EDUCATION

Lamar University | Bachelor of Science in Computer Game Development | Expected Summer 2027

Relevant CS coursework: Data Structures, Algorithm Design and Analysis, Operating Systems, Computer Architecture, Computer Networks, Computer Theory, Design of Programming Languages, Discrete Structures.

Mathematics: Linear Algebra, Calculus.

TECHNICAL SKILLS

Programming: C++, object-oriented programming, algorithm design

Software development: Debugging, software testing, state machines, code maintenance, version control

Tools and platforms: Visual Studio, Git, GitHub, Git LFS, Unreal Engine 5, Unity (introductory experience)

Additional concepts: Computer architecture, memory systems, gameplay systems, AI behavior logic

SELECTED PROJECTS

TPSCombatSandbox | Personal C++ / Unreal Engine 5 Project

- Built and tested a third-person interactive prototype featuring movement, aiming, weapons, and damage handling.
- Programmed state-driven enemy behaviors for detection, patrol, investigation, and combat; integrated animations and debugged interactions across systems.

Cache Simulator | Computer Architecture Coursework

- Implemented a program to simulate cache mapping and memory access behavior.
- Tested program outputs to better understand how memory addresses map to cache entries.

Dijkstra's Algorithm Program | Algorithms Coursework

- Worked on a shortest-path algorithm implementation and debugged program execution and output.

EXPERIENCE & ACTIVITIES

Independent Programming Projects: Develop and troubleshoot software projects outside of class; maintain source changes with Git and GitHub.

Academic activities: Programming assignments and technical coursework spanning algorithms, computing systems, and software development.

CERTIFICATIONS

No formal certifications currently held.